

Conditional Feature Importance revisited: Double Robustness, Efficiency and Inference

Angel REYERO LOBO

Global Sensitivity Analysis and Interpretability of Black-box Models

IMT & Inria Paris-Saclay

Joint work with:

Pierre NEUVIAL & Bertrand THIRION.

March 22, 2026



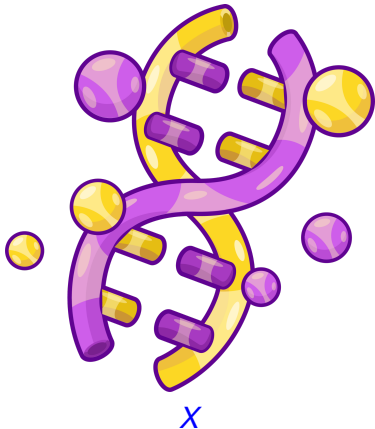
Contents

- 1 Introduction
- 2 Related work
- 3 Conditional Feature Importance revisited
 - Double Robustness of CFI
 - Sobol-CFI and efficiency
 - Inference
- 4 Experiments
- 5 Conclusion
- 6 References

- 1 Introduction
- 2 Related work
- 3 Conditional Feature Importance revisited
 - Double Robustness of CFI
 - Sobol-CFI and efficiency
 - Inference
- 4 Experiments
- 5 Conclusion
- 6 References

Motivation: Intrinsic Variable Importance

How can we define / learn the importance of each covariate X^j wrt an outcome y ?



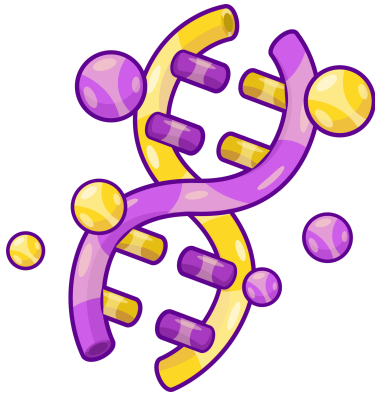
💡 Try to study their relationship using a ML model:



$$\hat{m} \in \operatorname{argmin}_{f \in \mathcal{F}} \hat{\mathbb{E}}[\mathcal{L}(f(X), y)].$$

Motivation: Intrinsic Variable Importance

How can we define / learn the importance of each covariate X^j wrt an outcome y ?



X

💡 Try to study their relationship using a ML model:



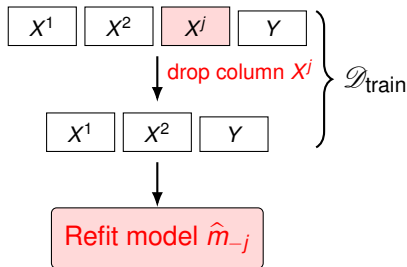
y

$$\hat{m} \in \operatorname{argmin}_{f \in \mathcal{F}} \hat{\mathbb{E}}[\mathcal{L}(f(X), y)].$$

- 1 Introduction
- 2 Related work
- 3 Conditional Feature Importance revisited
 - Double Robustness of CFI
 - Sobol-CFI and efficiency
 - Inference
- 4 Experiments
- 5 Conclusion
- 6 References

Model agnostic feature importance: Refitting vs Permuting

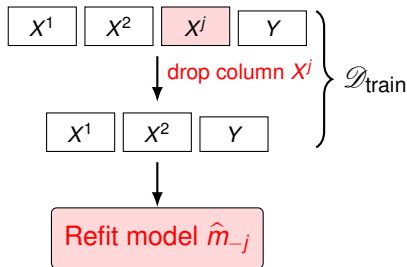
LOCO



$$\psi_{\text{LOCO}} = \text{perf}(\text{refit } \hat{m}_{-j}) - \text{perf}(\text{orig } \hat{m})$$

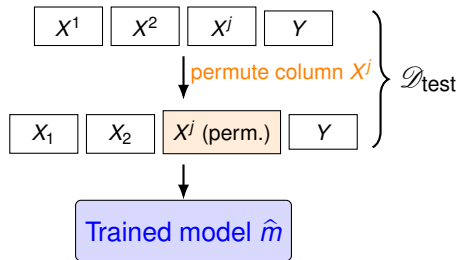
Model agnostic feature importance: Refitting vs Permuting

LOCO



$$\psi_{\text{LOCO}} = \text{perf}(\text{refit } \hat{m}_{-j}) - \text{perf}(\text{orig } \hat{m})$$

PFI



$$\psi_{\text{PFI}} = \text{perf}(\hat{m}(\text{perm } X^{(j)})) - \text{perf}(\hat{m}(\text{orig } X))$$

- **Leave One Covariate Out(LOCO):**

$$\hat{\psi}_{\text{LOCO}}(j, P_0) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathcal{L}(y_i, \hat{m}_{-j}(x_i^{-j})) - \mathcal{L}(y_i, \hat{m}(x_i)). \quad (1)$$

- ✓ It estimates an interpretable quantity ($\psi_{\text{TSI}}(j, P_0) = \mathbb{E} [\mathcal{L}(y, m_{-j}(X^{-j})) - \mathcal{L}(y, m(X))]$).
- ✓ Type-I error control ([Williamson et al. \(2021\) JASA](#)).
- ✗ In practice: unstable and computational intensive.

Literature review: General model-agnostic framework

- **Leave One Covariate Out(LOCO):**

$$\hat{\psi}_{\text{LOCO}}(j, P_0) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathcal{L}(y_i, \hat{m}_{-j}(x_i^{-j})) - \mathcal{L}(y_i, \hat{m}(x_i)). \quad (1)$$

- **Permutation Feature Importance(PFI):**

$$\hat{\psi}_{\text{PFI}}(j, P_0) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathcal{L}(y_i, \hat{m}(x_i^{(j)})) - \mathcal{L}(y_i, \hat{m}(x_i)). \quad (2)$$

where the j -th covariate is **permuted**.

- ✓ Fast (no need to retrain $\hat{m}$).
- ✗ Extrapolation bias ([Strobl et al. \(2007\) Bioinformatics](#)).
- ✗ Not interesting theoretically ([Bénard et al \(2022\) Biometrika](#)).
- 💡 Instead of breaking the relationship of X^j with X^{-j} and y , we only need to break it with y !

Literature review: General model-agnostic framework

- **Leave One Covariate Out(LOCO):**

$$\hat{\psi}_{\text{LOCO}}(j, P_0) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathcal{L}(y_i, \hat{m}_{-j}(\mathbf{x}_i^{-j})) - \mathcal{L}(y_i, \hat{m}(\mathbf{x}_i)). \quad (1)$$

- **Permutation Feature Importance(PFI):**

$$\hat{\psi}_{\text{PFI}}(j, P_0) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathcal{L}(y_i, \hat{m}(\mathbf{x}_i^{(j)})) - \mathcal{L}(y_i, \hat{m}(\mathbf{x}_i)). \quad (2)$$

where the j -th covariate is **permuted**.

- **Conditional Feature Importance(CFI):**

$$\hat{\psi}_{\text{CFI}}(j, P_0) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathcal{L}(y_i, \hat{m}(\tilde{\mathbf{x}}_i^{(j)})) - \mathcal{L}(y_i, \hat{m}(\mathbf{x}_i)), \quad (3)$$

where the j -th covariate is **conditionally permuted**.

✓ Works well in practice for feature selection ([Chamma et al. \(2023\) NeurIPS](#)).

✗ Not a theoretically interesting quantity, and its performance is not well understood.

- 1 Introduction
- 2 Related work
- 3 **Conditional Feature Importance revisited**
 - Double Robustness of CFI
 - Sobol-CFI and efficiency
 - Inference
- 4 Experiments
- 5 Conclusion
- 6 References

Improving variable selection: double robustness of CFI

- To detect a null j , it suffices to have either a good $\hat{m}$ or a good conditional sampler:

- If $\hat{m} \approx m \in \mathcal{F}_{-j} := \{f, f(u) = f(v) \text{ for } u_{-j} = v_{-j}\}$, then $m(\tilde{X}'^{(j)}) = m(X)$ and

$$\mathbb{E} [\mathcal{L}(y, m(\tilde{X}'^{(j)})) | \mathcal{D}_{\text{train}}] - \mathbb{E} [\mathcal{L}(y, m(X)) | \mathcal{D}_{\text{train}}] \approx 0.$$

- If $\tilde{X}'^{(j)} \approx \tilde{X}^{(j)}$ since $X^j \perp\!\!\!\perp y | X^{-j}$, then $\tilde{X}^{(j)} \stackrel{\text{i.i.d.}}{\sim} X$ and $\tilde{X}^{(j)j} \perp\!\!\!\perp y | X^{-j}$ so

$$\mathbb{E} [\mathcal{L}(y, \hat{m}(\tilde{X}^{(j)})) | \mathcal{D}_{\text{train}}] - \mathbb{E} [\mathcal{L}(y, \hat{m}(X)) | \mathcal{D}_{\text{train}}] \approx 0.$$

Improving variable selection: double robustness of CFI

- To detect a null j , it suffices to have either a good $\hat{m}$ or a good conditional sampler:

- If $\hat{m} \approx m \in \mathcal{F}_{-j} := \{f, f(u) = f(v) \text{ for } u_{-j} = v_{-j}\}$, then $m(\tilde{X}^{(j)}) = m(X)$ and

$$\mathbb{E} [\mathcal{L}(y, m(\tilde{X}^{(j)})) | \mathcal{D}_{\text{train}}] - \mathbb{E} [\mathcal{L}(y, m(X)) | \mathcal{D}_{\text{train}}] \approx 0.$$

- If $\tilde{X}^{(j)} \approx \tilde{X}^{(j)}$ since $X^j \perp\!\!\!\perp y | X^{-j}$, then $\tilde{X}^{(j)} \stackrel{\text{i.i.d.}}{\sim} X$ and $\tilde{X}^{(j)j} \perp\!\!\!\perp y | X^{-j}$ so

$$\mathbb{E} [\mathcal{L}(y, \hat{m}(\tilde{X}^{(j)})) | \mathcal{D}_{\text{train}}] - \mathbb{E} [\mathcal{L}(y, \hat{m}(X)) | \mathcal{D}_{\text{train}}] \approx 0.$$

Proposition 1

Assuming $y = X\beta + \varepsilon$ with X Gaussian and j null, then

$$\mathbb{E} [\hat{\psi}_{\text{LOCO}}(j, P_0)] = O(1/n_{\text{train}}),$$

$$\mathbb{E} [\hat{\psi}_{\text{CFI}}(j, P_0)] = O(1/n_{\text{train}}^2).$$

Sobol-CFI: a new Total Sobol Index estimate

🚩 **Goal:** $\psi_{\text{TSI}}(j, P_0) = \mathbb{E} [\mathcal{L}(y, m_{-j}(X^{-j})) - \mathcal{L}(y, m(X))].$

💡 We can use the Tower's property:

$$m_{-j}(X^{-j}) = \mathbb{E} [y \mid X^{-j}] = \mathbb{E} [\mathbb{E} [y \mid X] \mid X^{-j}] = \mathbb{E} [m(X) \mid X^{-j}] = \mathbb{E} [m(\tilde{X}^{(j)}) \mid X^{-j}].$$

Sobol-CFI: a new Total Sobol Index estimate

🚩 **Goal:** $\psi_{\text{TSI}}(j, P_0) = \mathbb{E} [\mathcal{L}(y, m_{-j}(X^{-j})) - \mathcal{L}(y, m(X))]$.

💡 We can use the Tower's property:

$$m_{-j}(X^{-j}) = \mathbb{E} [y \mid X^{-j}] = \mathbb{E} [\mathbb{E} [y \mid X] \mid X^{-j}] = \mathbb{E} [m(X) \mid X^{-j}] = \mathbb{E} [m(\tilde{X}^{(j)}) \mid X^{-j}].$$

- Generate n_{cal} conditionally independent samples/ observation.
- **Sobol-CFI:**

$$\hat{\psi}_{\text{Sobol-CFI}}(j, P_0) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathcal{L} \left(y_i, \frac{1}{n_{\text{cal}}} \sum_{l=1}^{n_{\text{cal}}} \hat{m}(\tilde{x}_{i,l}^{(j)}) \right) - \mathcal{L}(y_i, \hat{m}(x_i)).$$

- ✓ It is **nonparametric efficient**, **fast**, and **stable**.
- ✓ It links removal with permutation approaches (**explainable**).

Sobol-CFI: a new Total Sobol Index estimate

🚩 **Goal:** $\psi_{\text{TSI}}(j, P_0) = \mathbb{E} [\mathcal{L}(y, m_{-j}(X^{-j})) - \mathcal{L}(y, m(X))]$.

💡 We can use the Tower's property:

$$m_{-j}(X^{-j}) = \mathbb{E} [y \mid X^{-j}] = \mathbb{E} [\mathbb{E} [y \mid X] \mid X^{-j}] = \mathbb{E} [m(X) \mid X^{-j}] = \mathbb{E} [m(\tilde{X}^{(j)}) \mid X^{-j}].$$

- Generate n_{cal} conditionally independent samples/ observation.
- **Sobol-CFI:**

$$\hat{\psi}_{\text{Sobol-CFI}}(j, P_0) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathcal{L} \left(y_i, \frac{1}{n_{\text{cal}}} \sum_{l=1}^{n_{\text{cal}}} \hat{m}(\tilde{x}_{i,l}^{(j)}) \right) - \mathcal{L}(y_i, \hat{m}(x_i)).$$

- ✓ It is **nonparametric efficient**, **fast**, and **stable**.
- ✓ It links removal with permutation approaches (**explainable**).
- If $\mathcal{L} = \ell^2$, fixing n_{cal} , then $\hat{\psi}_{\text{Sobol-CFI}}(j, P_0) \xrightarrow{n_{\text{train}}, n_{\text{test}} \rightarrow \infty} (1 + 1/n_{\text{cal}}) \psi_{\text{TSI}}(j, P_0)$.

Algorithm 1 Nonparametric Sobol-CFI (Sobol-CFI-ST/Wilcox)

Input: Model $\hat{m}$, test data $\{(X_i, y_i)\}_i^{n_{\text{test}}}$, control level α and a feature j

(Regression) Fit $\hat{v}_j \approx \mathbb{E}[X^j \mid X^{-j}]$

Compute the residuals $\hat{\varepsilon}_j = X^j - \hat{v}_j(X^{-j})$

Construct

$$\tilde{X}'^{(j)} = \hat{v}_j(X^{-j}) + \hat{\varepsilon}_j^{\text{permuted}}$$

Compute nonparametric paired test (Sign-test/Wilcoxon) between

$$\left\{ \mathcal{L}(\hat{m}(\mathbf{X}_i), y_i) \right\}_i^{n_{\text{test}}} \text{ and } \left\{ \mathcal{L}(\hat{m}(\tilde{X}_i'^{(j)}), y_i) \right\}_i^{n_{\text{test}}}$$

Return the computed p-value p'

- Type-I error control: $\mathbb{P}_{\mathcal{H}_0}(p' \leq \alpha \mid \{X_i^{-j}, y_i\}_i^{n_{\text{test}}}, \mathcal{D}_{\text{train}}) \leq \alpha + d_{\text{TV}}(\tilde{X}'^{(j)}, \tilde{X}^{(j)})$.

- 1 Introduction
- 2 Related work
- 3 Conditional Feature Importance revisited
 - Double Robustness of CFI
 - Sobol-CFI and efficiency
 - Inference
- 4 Experiments**
- 5 Conclusion
- 6 References

Double robustness in practice

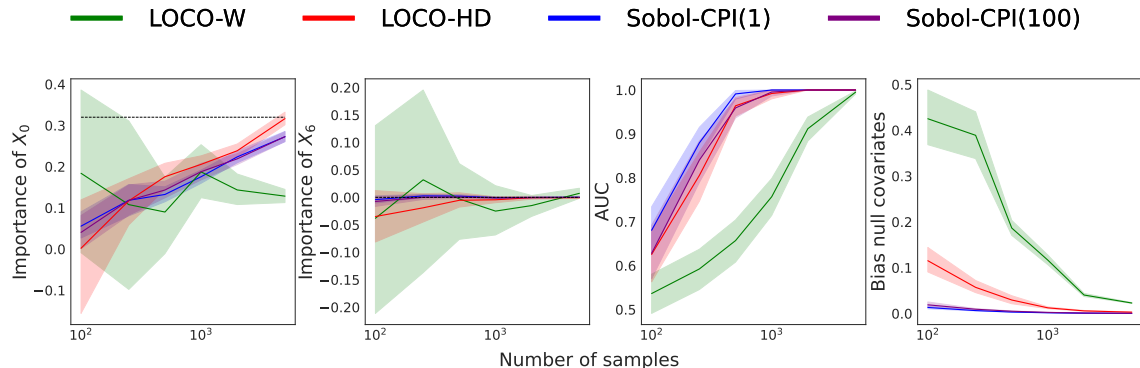


Figure 1: Double robustness with complex learners: $\hat{m}$ a Superlearner and $y = X_0 X_1 \mathbb{I}_{X_2 > 0} + 2X_3 X_4 \mathbb{I}_{X_2 < 0}$, with $X \sim \mathcal{N}(\mu, \Sigma)$, where $\Sigma_{i,j} = 0.6^{|i-j|}$, $p = 50$, and $\mu = \mathbf{0}$.

Double robustness and inference on real data

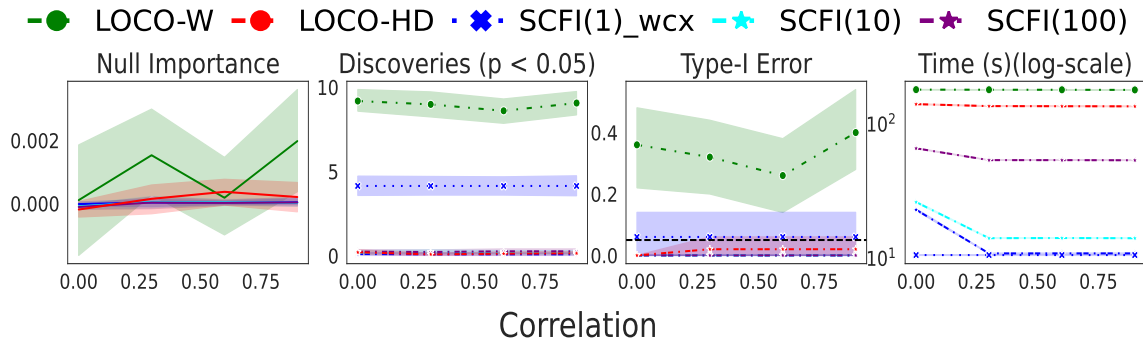


Figure 2: Double robustness and inference on real data: Using WBDC dataset with an added artificial null feature of varying correlation (x-axis). SCFI-based methods assign no importance to the null. With Wilcoxon test, discoveries are made while controlling the error and remaining computationally efficient.

- 1 Introduction
- 2 Related work
- 3 Conditional Feature Importance revisited
 - Double Robustness of CFI
 - Sobol-CFI and efficiency
 - Inference
- 4 Experiments
- 5 Conclusion**
- 6 References

We revisited the CFI, including the estimation of the **conditional sampler**:

- ✓ A **double robustness** property:
To detect a null, it suffices to have either a good $\hat{m}$ or a good **conditional sampler**.
- ✓ Sobol-CFI: an efficient estimator of the **Total Sobol Index**.
- ✓ A valid and consistent **type-I error** test.

Conclusion

Methods	LOCO	PFI	CFI	Sobol-CFI
Fast	✗	✓	✓	✓
No extrapolation	✓	✗	✓	✓
Interpretable	✓	✗	✗	✓
Double Robustness	✗	✗	✓	✓
Type-I error control	✓	✗	✗	✓

- 1 Introduction
- 2 Related work
- 3 Conditional Feature Importance revisited
 - Double Robustness of CFI
 - Sobol-CFI and efficiency
 - Inference
- 4 Experiments
- 5 Conclusion
- 6 References

References

- Clément Bénard, Sébastien Da Veiga, and Erwan Scornet. Mean decrease accuracy for random forests: inconsistency, and a practical solution via the sobol-mda. *Biometrika*, 109(4):881–900, 02 2022.
- Ahmad Chamma, Denis A Engemann, and Bertrand Thirion. Statistically valid variable importance assessment through conditional permutations. *Advances in Neural Information Processing Systems*, 36, 2024.
- Angel Reyero-Lobo, Pierre Neuvial, and Bertrand Thirion. A principled approach for comparing variable importance, 2025. URL <https://arxiv.org/abs/2507.17306>.
- Angel Reyero-Lobo, Pierre Neuvial, and Bertrand Thirion. Conditional feature importance revisited: Double robustness, efficiency and inference, 2026. URL <https://arxiv.org/abs/2501.17520>.

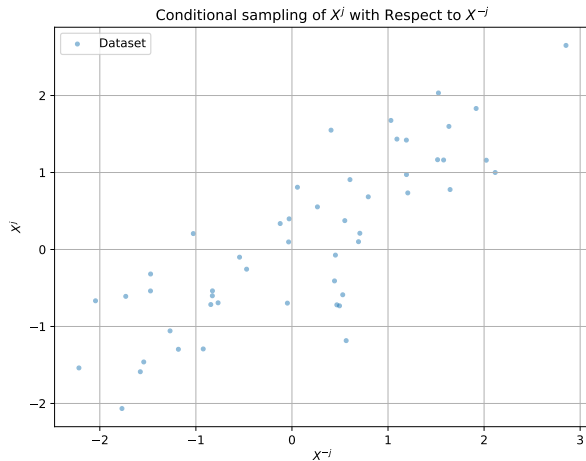
- Carolin Strobl, Anne-Laure Boulesteix, Thomas Kneib, Thomas Augustin, and Achim Zeileis. Conditional variable importance for random forests. *BMC Bioinformatics*, 9(1): 307, 2008. ISSN 1471-2105. doi: 10.1186/1471-2105-9-307. URL <https://doi.org/10.1186/1471-2105-9-307>.
- Brian D Williamson, Peter B Gilbert, Noah R Simon, and Marco Carone. A general framework for inference on algorithm-agnostic variable importance. *Journal of the American Statistical Association*, 118(543):1645–1658, 2023.

Thank You, Questions?

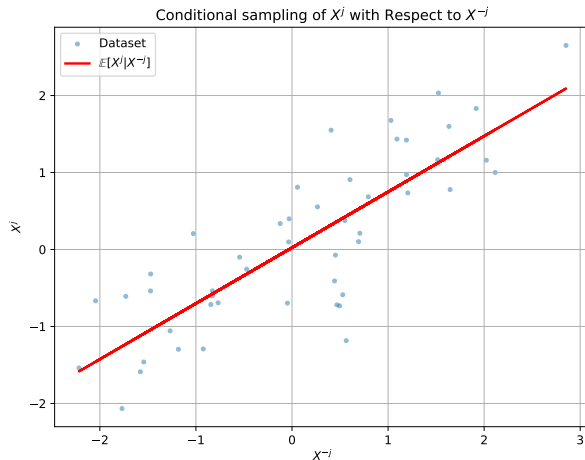
7 Conditional permutation

8 Additional experiments

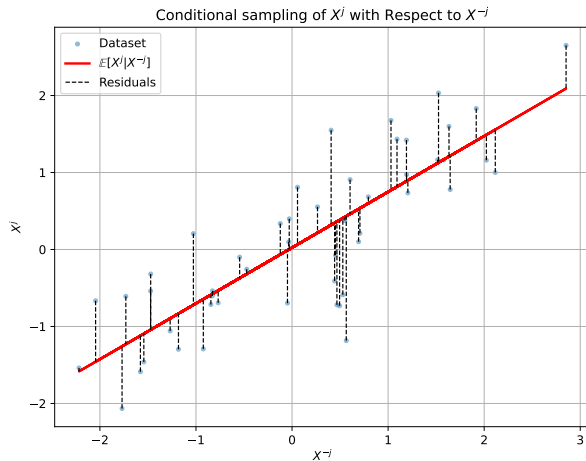
An idea to permute conditionally on X^{-j}



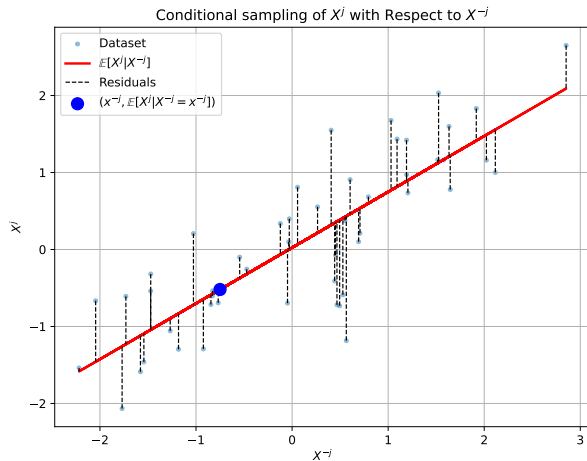
An idea to permute conditionally on X^{-j}



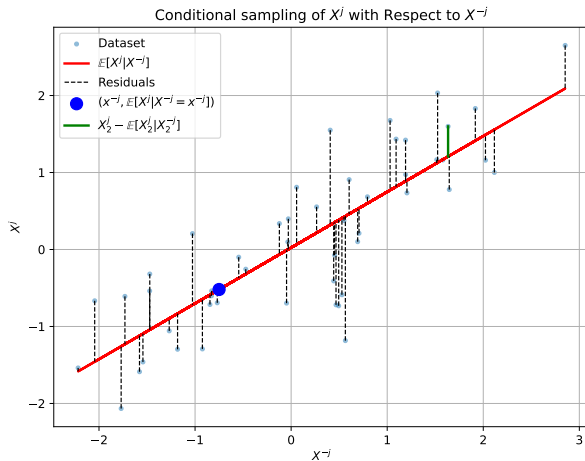
An idea to permute conditionally on X^{-j}



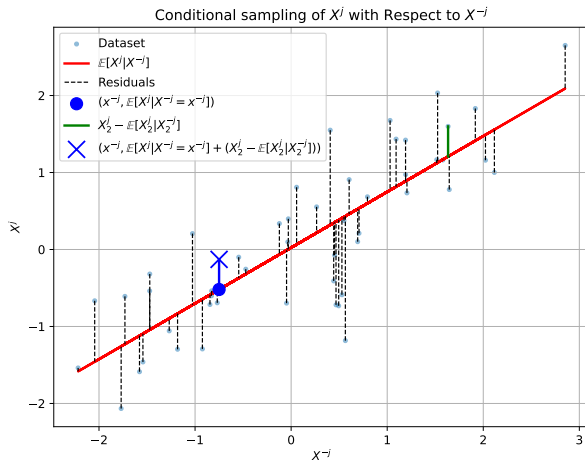
An idea to permute conditionally on X^{-j}



An idea to permute conditionally on X^{-j}



An idea to permute conditionally on X^{-j}



Validity of the conditional sampling

- In practice, we need to train a regressor $\hat{v}_{-j}$ of X^j on X^{-j} . Then, for an x , we predict $\hat{v}_{-j}(x^{-j})$ and add a permuted residual $(x'^j - \hat{v}_{-j}(x'^j))$.

Assumption 1

$X^j = v_{-j}(X^{-j}) + \varepsilon$ with $\varepsilon \perp\!\!\!\perp X^{-j}$.

Lemma 2 (Sampling validity)

Under Assumption 1 and assuming the consistency of $\hat{v}_{-j}$, the conditional step of the CPI, presented in [Chamma et al.\(2023\) NeurIPS](#), is valid.

7 Conditional permutation

8 Additional experiments

Different samplers and asymptotic relevance

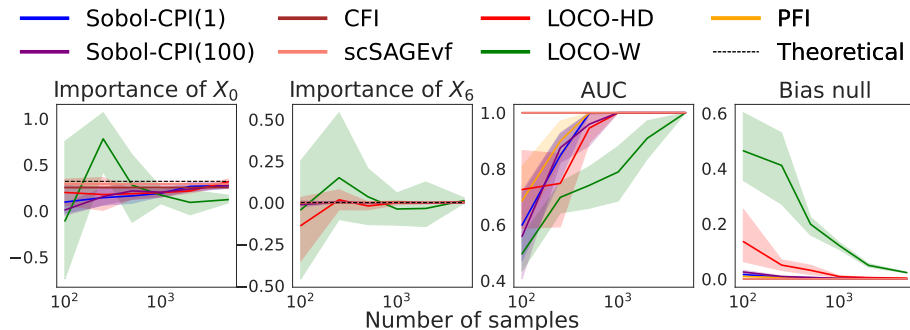


Figure 3: Double robustness for complex learners: left to right: TSI estimates for an important covariate (X_0) and a null covariate (X_6); AUC for an importance-based variable selection; bias for null covariates.

Effect of n-cal

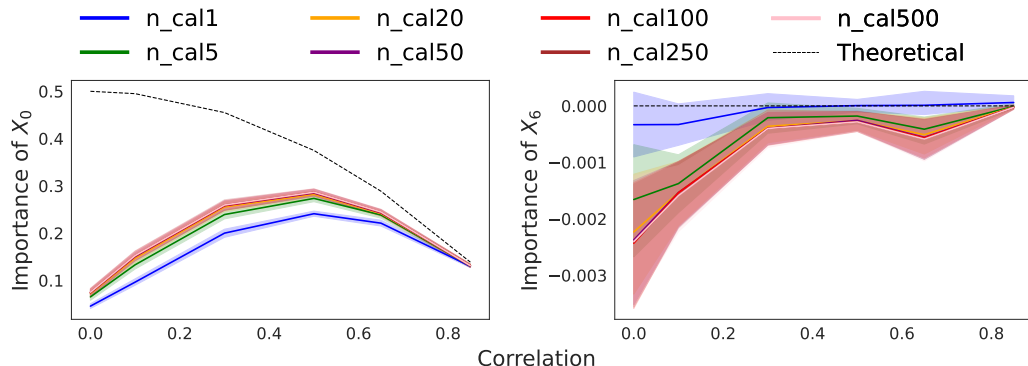


Figure 4: Calibration set size effect as a trade-off between Variable Importance and Variable Selection: TSI estimation in a nonlinear setting. On the left, an important covariate (X_0), and on the right a non-important covariate (X_6). With a larger n_{cal} , the importance estimation of the non-null covariate is slightly improved, enhancing variable importance. For the null, there is a slightly greater bias, making variable selection less accurate.